

Demonstration of Closed Loop AI-Driven RAN Controllers Using O-RAN SDR Testbed

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Abstract—Open Radio Access Network (O-RAN), a virtualized, modular, and disaggregated design paradigm for 5G/NextG cellular RANs, aims to integrate intelligence into cellular networks, enabling advanced deployment, operation and maintenance of the network. A key component of O-RAN is the RAN Intelligent Controller (RIC), which facilitates the controllability of RAN elements through data-driven, closed-loop, intelligent control using software microservices called extended Applications (xApps). In this demonstration, we showcase intelligent RAN control using a custom *interference classification xApp* hosted within the near-real-time RIC. This demonstration provides a tangible example for designing, testing, and experimenting with AI-driven RAN controllers and exploring various possibilities within the O-RAN system.

Keywords—5G, O-RAN, xApp, intelligent RAN control

I. INTRODUCTION

Open Radio Access Network (O-RAN) is transforming the cellular wireless technology from inflexible and monolithic systems to more flexible, modular, interoperable and disaggregated networks[1]. This allows network operators to mix and match hardware and software components from different vendors, resulting in increased flexibility, scalability, and cost savings for network operators. Furthermore, a new O-RAN component, called the RAN Intelligent Controller (RIC), allows monitoring and controlling of various RAN functionalities. It consists of two parts: a non-real-time RIC that handles high-level orchestration and management of RAN, operating at $> 1s$ intervals, and a near-real-time (near-RT) RIC that provides real-time (10ms - 1s) monitoring and control of RAN, such as resource scheduling and interference classification. Third party microservices, namely, extended applications (xApps) can be hosted within the near-RT RIC, which may collect data from the RAN (e.g., I/Q samples), process it, and send back control actions to the RAN.

In this work, we demonstrate AI-driven RAN control using a custom *interference classification xApp*, hosted in the near-real-time RIC of the O-RAN architecture. To prototype this, we developed and tested our xApp's closed-loop functionality using a software-defined radio (SDR)-based O-RAN testbed.

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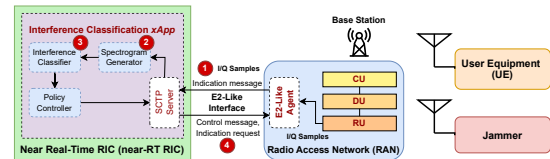


Fig. 1: Procedure details of Interference classification xApp

Our custom xApp employs a machine learning model to detect interference in spectrograms generated from collected I/Q samples via an open interface. The model's predictions are then transmitted to the RAN via the same interface, allowing the RAN to respond by taking appropriate measures, such as modulation and coding (MCS) selection, to mitigate interference and enhance network performance.

II. O-RAN SYSTEM DESIGN AND IMPLEMENTATION

We have designed and implemented a fully open-source O-RAN testbed platform utilizing software-defined radios (SDRs). Our O-RAN SDR testbed incorporates an LTE network that seamlessly interacts with the near-real-time RIC, along with its hosted xApps through open interfaces.

To construct our LTE network, we employ the srsRAN[2] 4G software suite, an open-source cellular software solution. This suite provides essential components such as an LTE core, a 4G base station, and user equipment (UE) to emulate a functional LTE network. As for the near-RT RIC, we leverage the E-Release version provided by the open-source O-RAN Software Community[3]. When establishing communication between the LTE network and the near-RT RIC, we utilize an E2-like interface. This interface offers similar functionalities as the O-RAN compliant E2 interface but simplifies the subscription process, as well as message encoding and structure. To enable seamless communication via the E2-like interface, we develop a modified version of srsRAN that incorporates an E2-Like Agent within the eNodeB. This agent acts as an intermediary, facilitating smooth communication between the near-RT RIC and the RAN. Finally, we develop an interference classification xApp, which leverages the E2-like interface to exert control over the RAN.

The stepwise procedure of the interference classification xApp visualized in Fig. 1, is as follows:

- 1) The RAN collects I/Q samples and sends them to the xApp through the E2-like interface. For our demonstration, we periodically collect 10 ms segments of I/Q data, equivalent to the length of one LTE frame.
- 2) The Spectrogram Generator takes the I/Q samples received from the E2-like interface and generates a spectrogram, which can be used as the input for an ML model developed using CNNs.
- 3) The Interference Classifier processes the generated spectrograms using the CNN model. The classifier outputs the predicted type of interference present in the signal.
- 4) The classifier's prediction is sent to the Policy Controller, where a decision to control the RAN is made. These decisions are communicated via control messages from the xApp to the eNB through the E2-like interface, and allow us to update the BS configuration. For our demonstration, when interference is detected, the MCS is automatically adjusted. srsRAN is capable of dynamically adjusting the MCS based on the signal's condition, so we enable this functionality when interference is detected and disable it (setting the MCS to a fixed value) when interference is no longer detected.

O-RAN testbed: As shown in Fig. 2, the srsEPC core and srsRAN eNB are hosted on the same workstation, which is connected to a USRP B210 SDR for over-the-air communications. On a separate workstation, we have the srsRAN UE, which is configured to connect to the LTE network using another B210. The jammer module is hosted on a third workstation and is also connected to a B210. Using MATLAB, the jammer generates interference signals, which are then transmitted to interfere with the communication link between the eNB and UE. The near-RT RIC (and xApp) is deployed on a virtual machine hosted on a rack server, which is connected via ethernet to the eNB.

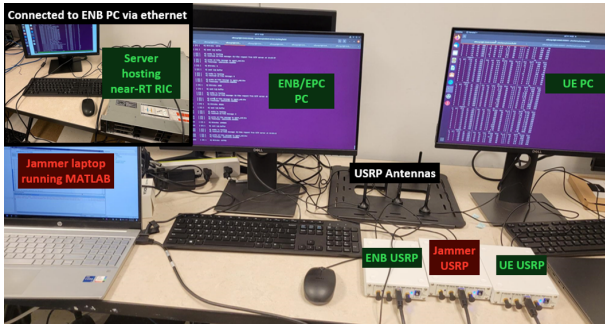


Fig. 2: O-RAN SDR testbed

III. INTERFERENCE CLASSIFICATION xAPP DEMONSTRATION

For demonstration purposes, we consider the following scenario: The UE runs iPerf3 and regularly sends data traffic to eNB using LTE uplink signal (as SOI), while the jammer attempts to interfere the SOI with either transmitting CI or CWI interfering signal in the same uplink frequency. The scalograms generated by these signals are shown in Fig. 3. As

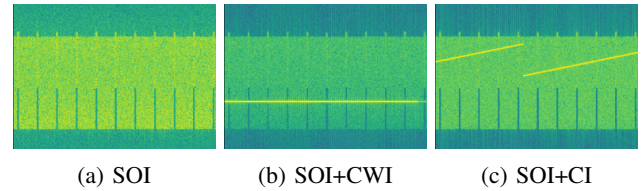


Fig. 3: Scalograms based on collected I/Q data

discussed in Section II and shown in Fig. 1, the interference classification using an AI-driven xApp follows the four steps, attempting to detect and mitigate the interference in real-time, and improve network performance.

Fig. 4 shows the uplink bitrate of the LTE network under two scenarios (i) Under interference (no RAN control via xApp), (ii) Under interference (with RAN control via xApp). The xApp is able to detect the interference, adjust MCS accordingly, and thereby, improving the network performance.

Dataset and Model training: The dataset used to train our model was collected using the O-RAN testbed shown in Fig. 2. We collected 1200 samples for each class and our model was built using a CNN similar to the parameters and complexity found in [4]. The model was trained offline before it was deployed as an xApp in the near-RT RIC.

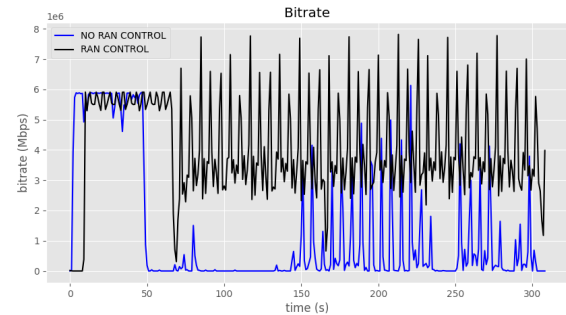


Fig. 4: Uplink bitrate vs. time

CONCLUSION

The demonstration featured a closed-loop, AI-driven RAN control utilizing an example xApp on an O-RAN SDR testbed. The software employed, including the customized srsRAN, near-RT RIC, and E2-like xApp, are now available as part of the Open AI Cellular platform [5].

REFERENCES

- [1] "O-RAN Alliance e.V." (2023), [Online]. Available: <https://www.o-ran.org/>.
- [2] Software Radio Systems. "srsRAN Project." (2023), [Online]. Available: <https://www.srslte.com/>.
- [3] "O-RAN Software Community." (2023), [Online]. Available: <https://wiki.o-ran-sc.org/display/ORAN>.
- [4] T. Oyedare, V. K. Shah, D. J. Jakubisin, and J. H. Reed, "Keep It Simple: CNN Model Complexity Studies for Interference Classification Tasks," *IEEE INFOCOM Workshop*, 2023.
- [5] "Open AI Cellular." (2023), [Online]. Available: <https://github.com/openaicellular>.